Lab No 4

Question 1

Create a query to display the **employee name** and the **name of the department** for each employee **in UPPERCASE**.
Make sure all other columns retain their original case formatting.

```sql
SELECT
    UPPER(e.first_name || ' ' || e.last_name) AS EMPLOYEE_NAME,
    UPPER(d.department_name) AS DEPARTMENT_NAME
FROM employees e
JOIN departments d
ON e.department_id = d.department_id;
```

Script Output × | Query Result ×

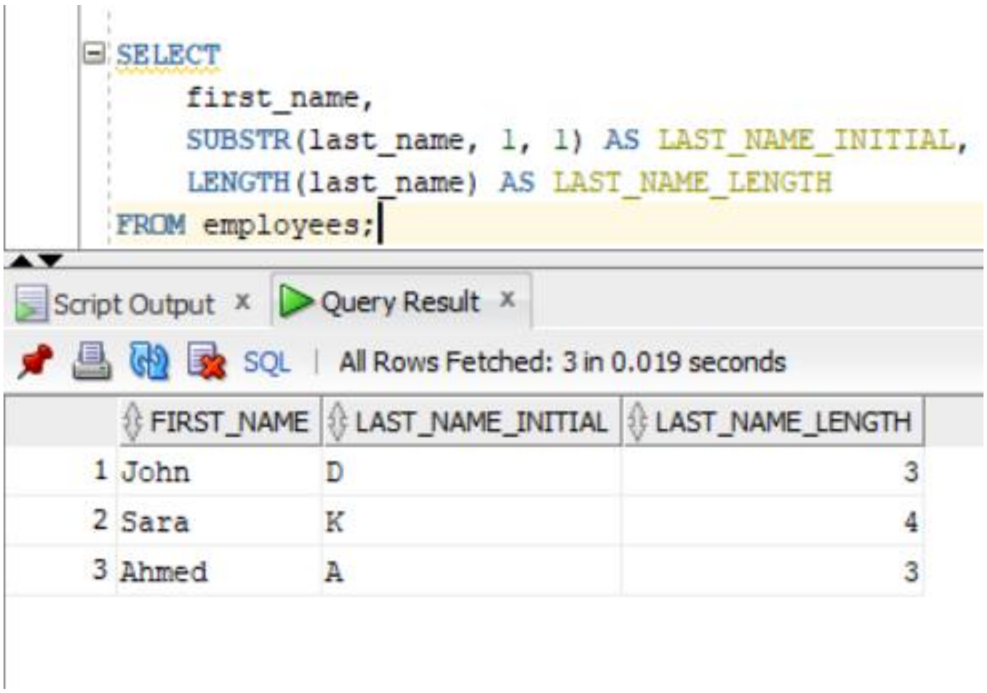
SQL | All Rows Fetched: 3 in 0.004 seconds

	EMPLOYEE_NAME	DEPARTMENT_NAME
1	JOHN DOE	HUMAN RESOURCES
2	SARA KHAN	FINANCE
3	AHMED ALI	IT

Question 2

Create a query to find the **length of each employee's last name**. Display the results in **three columns**:

1. First Name
2. First letter of Last Name
3. Length of the employee’s Last Name

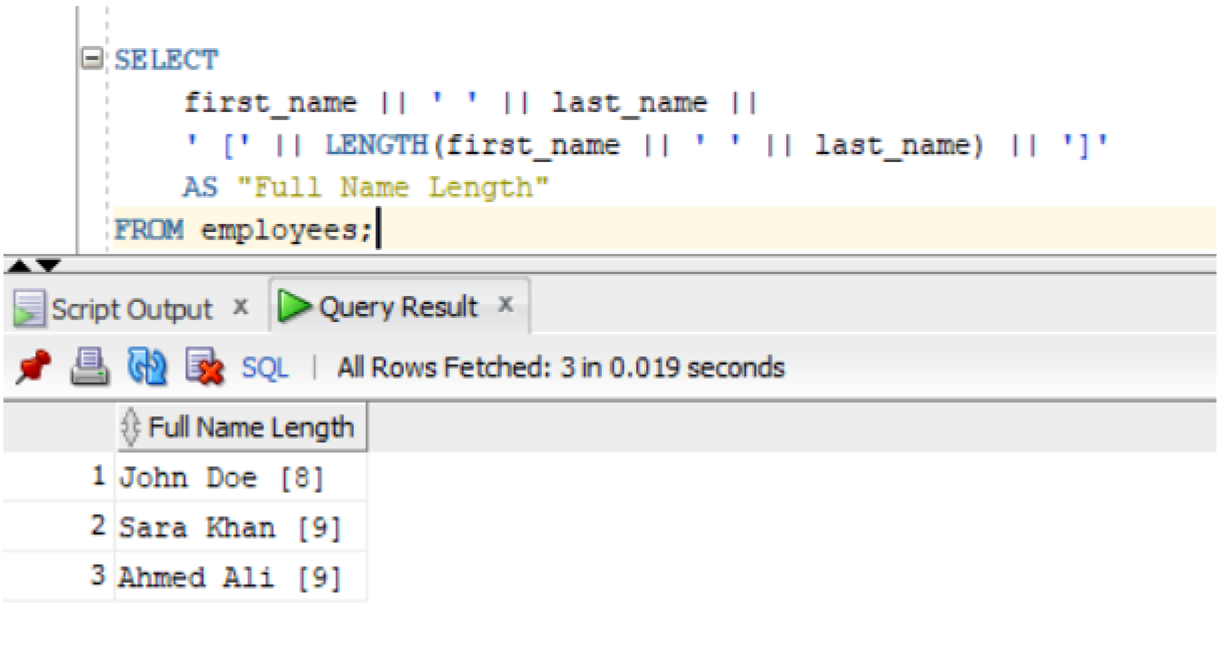


```sql
SELECT
    first_name,
    SUBSTR(last_name, 1, 1) AS LAST_NAME_INITIAL,
    LENGTH(last_name) AS LAST_NAME_LENGTH
FROM employees;
```


	FIRST_NAME	LAST_NAME_INITIAL	LAST_NAME_LENGTH
1	John	D	3
2	Sara	K	4
3	Ahmed	A	3

Question 3

Write a query to display employee names along with the length of their full names.
Also concatenate the results in the format:
“John Doe [7]”
Label the column as **Full Name Length**



Question 4

Extract the **first 3 characters of each employee's first name**.

```sql
SELECT
    first_name,
    SUBSTR(first_name, 1, 3) AS FIRST_3_CHARS
FROM employees;
```

Script Output | Query Result

SQL | All Rows Fetched: 3 in 0.006 seconds

	FIRST_NAME	FIRST_3_CHARS
1	John	Joh
2	Sara	Sar
3	Ahmed	Ahm

Question 5

Write a query to display product names and **rounded product prices to nearest integer**, ordered by rounded price in **descending order**.

```sql
SELECT
    product_name,
    ROUND(price) AS rounded_price
FROM products
ORDER BY ROUND(price) DESC;
```

Script Output | Query Result

SQL | All Rows Fetched: 4 in 0.021 seconds

	PRODUCT_NAME	ROUNDED_PRICE
1	Laptop	1051
2	Monitor	300
3	Keyboard	79
4	Mouse	25

Question 6

Display product prices **rounded to the nearest tens place**.

```sql
SELECT
    product_name,
    price,
    ROUND(price, -1) AS rounded_to_tens
FROM products;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 4 in 0.008 seconds

	PRODUCT_NAME	PRICE	ROUNDED_TO_TENS
1	Laptop	1050.75	1050
2	Mouse	25.49	30
3	Keyboard	78.99	80
4	Monitor	299.6	300

Question 7

Find the remainder when product price is divided by 5.
Display:

- Rounded price
- Truncated price
  Label columns as:
- ROUNDED_PRICE
- TRUNCATED_PRICE

```sql
SELECT
    product_name,
    ROUND(price) AS ROUNDED_PRICE,
    TRUNC(price) AS TRUNCATED_PRICE
FROM products;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 4 in 0.018 seconds

	PRODUCT_NAME	ROUNDED_PRICE	TRUNCATED_PRICE
1	Laptop	1051	1050
2	Mouse	25	25
3	Keyboard	79	78
4	Monitor	300	299

Question 8

Display the **order date truncated to the start of the month**.

```sql
SELECT
    order_id,
    order_date,
    TRUNC(order_date, 'MM') AS start_of_month
FROM orders;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.012 seconds

	ORDER_ID	ORDER_DATE	START_OF_MONTH
1	1	10-APR-26	01-APR-26
2	2	25-MAR-26	01-MAR-26
3	3	23-APR-26	01-APR-26

Question 9

Concatenate employee name and rounded salary to nearest 100.
Display result in format:
“Doe, John | Salary 2000”
Label column as **EMP_PAY_DETAILS**

```sql
SELECT
    last_name || ', ' || first_name ||
    ' | Salary ' || ROUND(salary, -2) AS EMP_PAY_DETAILS
FROM employees;
```

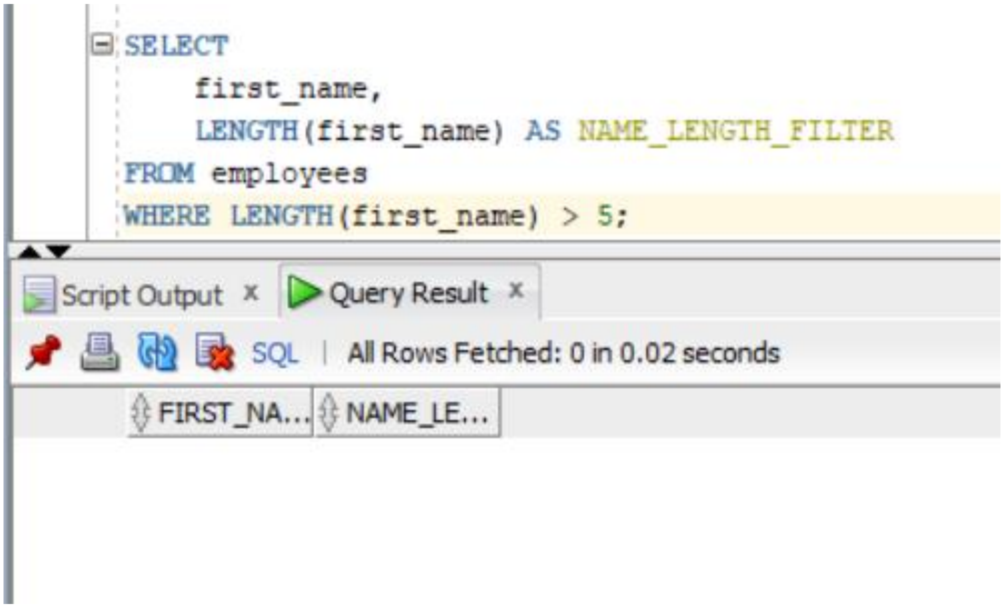
Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.02 seconds

	EMP_PAY_DETAILS
1	Doe, John | Salary 2000
2	Khan, Sara | Salary 3500
3	Ali, Ahmed | Salary 4200

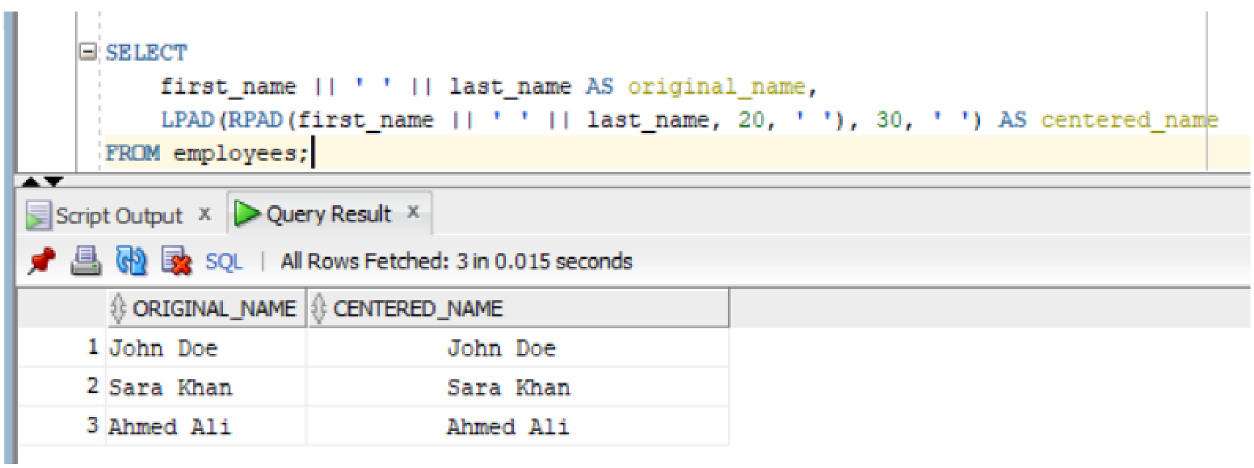
Question 10

Write a query to find employees where the **length of first name is greater than 5**.
Label the column as **NAME_LENGTH_FILTER**



Question 11

Write a query to display employee names **justified in the middle** of a fixed width column using **LPAD and RPAD**.



Question 12

Write the differences between ROUND and TRUNC. OR Write the differences between INSTR and SUBSTR.

Difference between ROUND and TRUNC

ROUND and TRUNC are both used for working with numbers in Oracle, but they behave differently.

ROUND is used to round a number to the nearest value based on normal mathematical rules. If the digit after the rounding point is 5 or more, it increases the value, otherwise it keeps it the same. For example, ROUND(78.6) becomes 79 and ROUND(78.4) becomes 78.

TRUNC, on the other hand, simply removes the decimal part without rounding. It does not consider the value after the decimal point. For example, TRUNC(78.6) becomes 78 and TRUNC(78.9) also becomes 78.

So, the main difference is that ROUND adjusts the value based on rules, while TRUNC just cuts off the decimal part without any rounding.

Difference between INSTR and SUBSTR

INSTR and SUBSTR are both string functions in Oracle, but they are used for different purposes.

INSTR is used to find the position of a character or substring inside a string. It returns a numeric value showing where the character is located. For example, INSTR('John Doe', 'D') returns 6 because 'D' starts at position 6.

SUBSTR is used to extract a part of a string from a given position. It returns the actual text instead of the position. For example, SUBSTR('John Doe', 1, 4) returns 'John'.

So, the main difference is that INSTR tells the position of a character, while SUBSTR extracts a portion of the string.